8

Applying Cognitive Theory in Public Health Investigations: Enhancing Food Recall with the Cognitive Interview

RONALD P. FISHER
and KATHRYN L. QUIGLEY

Public-health epidemiologists rely heavily on histories of food consumption to investigate foodborne outbreaks. Those histories, usually obtained between two to seven days after the suspect meal, are used to identify specific foods consumed by ill and well persons present at the implicated meal, to determine which foods are associated with the illness (Bryan, 1973). If a laboratory examination of the foods served at the meal is impossible or inadequate, the food-consumption histories may be the only source of information available to link a foodborne outbreak with a specific food (Mann, 1981). In such cases it is particularly important to obtain accurate food-consumption histories. The few studies conducted to examine the accuracy of such histories, however, indicate that errors in food-consumption recall are frequent (Decker, Booth, Dewey, Fricker, Hutcheson, and Schaffner, 1986; Mann, 1981). Furthermore, it has been suggested that there is little evidence or optimism that these food-consumption histories can be improved by employing various interview techniques (Decker et al., 1986).

The goal of the current research was to improve food-consumption histories by applying a variant of the *cognitive interview*, a technique that has been found to enhance respondent recollection in another discipline: eyewitness memory for the details of a crime (e.g., Fisher, Geiselman, Raymond, Jurkevich, and Warhaftig, 1987; Fisher, Geiselman, and Raymond, 1987; Geiselman, Fisher, MacKinnon, and Holland, 1985). The cognitive interview is based on principles of memory retrieval, cognition, and communication, as gathered from laboratory research in cognitive psychology and from extensive analysis of tape-recorded interviews conducted in the field (Fisher, Geiselman, and Raymond, 1987). Although the cognitive interview was developed initially to enhance witness recollection in criminal investigations, with minor modifications it should be applicable to other types of investigative interviewing, since it is based primarily on general principles of cognition. The present study examined the efficacy of the technique to obtain more extensive and accurate food-consumption histories in a simulated foodborne-outbreak investigation.

The theoretical framework of the cognitive interview, as applied to eliciting food-consumption histories, is based on five general principles of cognition and memory retrieval: context reinstatement, focused retrieval, extensive retrieval, varied retrieval, and multiple representations. Below we provide a simple conceptual description of each principle. A more detailed, concrete, step-by-step description for conducting investigative interviews is provided elsewhere (Fisher and Geiselman, in press).

The principle of context reinstatement suggests that an event will be better recalled if the rememberer is in the same psychological environment as when the event occurred originally (Tulving and Thomson, 1973). Thus, the interviewer encourages the respondent to think about the environmental context (e.g., dining-room arrangement, lighting conditions) and the relevant psychological context (e.g., why the respondent selected specific foods) at the time of the original meal. The principle of focused retrieval is based on the concept that memory retrieval, especially for details, requires mental concentration (e.g., Kahneman, 1973). Any distractions from this mental concentration, whether physical (e.g., extraneous noise) or psychological (e.g., interrupting the respondent's narration to ask a question), will disrupt memory retrieval. The principle of extensive retrieval states that the more retrieval attempts one makes, the more successful recall will be (Roediger and Payne, 1982). In practice this principle is fulfilled by encouraging respondents to search through memory even if they indicate that they have recalled as much as possible. It is important to note that when the respondent is encouraged to make additional searches through memory, the interviewer cannot simply ask the same question as posed originally. Such an approach often leads the respondent to indicate, "I already told you, I don't know." The interviewer must vary the form of the question, at least superficially, so that the respondent is induced to make another search through memory. The concept of varied retrieval is based on the notion that memories not activated by one retrieval probe may be accessed with another probe (Anderson and Pichert, 1978). Thus, a respondent who cannot recall a fact when asked a direct question (e.g., "Did you eat any vegetables?") may provide the answer when asked a different question ("What foods did you select last?"). In general changing the dimension of the question (e.g., from categorical to temporal) is likely to elicit new information. The final principle, multiple representations, suggests that the event to be remembered is stored in several forms (Fisher and Chandler, 1984). For example, the foods eaten are stored in several "mental images." There may be one image of the food while on the plate, another image of the food being served by the waiter, a third image of the food being carried away after the meal, and so on. Each of these images should be evoked and probed separately, exploiting all of the contents before probing other images (Fisher and Quigley, 1988). In order to be aware of the respondent's multiple representations of the meal, at the beginning of the interview, the interviewer should encourage the respondent to provide an open-ended description of all of the events related to the meal in question. During this initial narration the interviewer should note the various mental representations the respondent has of the meal and probe them later.

In a pilot study of food recall, more accurate food-consumption histories were obtained by using the cognitive interview than simply asking participants to recall the foods they had eaten at an earlier meal. In that study, nine students were encouraged to sample party foods (e.g., chips, beverages, cookies) from a buffet spread containing seventeen items. The students were told that they were participating in a study about how people eat and were not made aware of a forthcoming memory test. Two to seven days later, the students were interviewed for their recollections of the foods they had eaten. Four students were given a "standard" interview and were asked to recall the foods they had selected and also those not selected, but which were available from the table. The remaining five students were given the cognitive interview.

The cognitive interview elicited approximately 20% to 25% more correct information than did the standard interview. This advantage held both for foods eaten and those available but not eaten. Furthermore, the cognitive interview generated fewer incorrectly recalled foods. Although the sample was too small to produce conventional levels of statistical reliability, the magnitude of the effect was approximately the same as that found in the earlier eyewitness studies. The present study was conducted, in part, to extend the results of the pilot study to a larger sample of participants.

In foodborne-outbreak investigations, a menu of the items served at the suspect meal is occasionally available for review to facilitate the respondent's recollection. Does the presence of a menu nullify the advantage conferred by the cognitive interview? As a check on this possibility, and to increase the ecological validity of the research, the current study employed a recognition test (comparable to selecting from a menu) in addition to the recall task (generating the foods).

Method

Subjects

Twenty-six undergraduate students from Florida International University participated voluntarily in the experiment. The subjects were assigned randomly to one of two conditions, resulting in thirteen subjects per condition.

Procedure

Shortly after completing an in-class examination, the students were invited to another classroom to eat party foods. Those students who wished to participate voluntarily arrived at the "dining room" at various times, depending upon when they finished the examination. Upon arrival the students were encouraged to sample from the foods available. Thirty-four different foods (soft beverages, chips, pretzels, nuts, crackers, cheeses, fruits, raw vegetables, dips, pickles, olives, cookies, and candies) were displayed on two adjacent buffet tables. Plates and utensils were provided and seating was available in the "dining room," so the subjects could sit and converse casually with one another while eating. The subjects were permitted to return to the tables to select additional foods as often as they desired. Unlimited time was available for subjects to eat as much as they wished before leaving. Typical eating times were fifteen to twenty minutes.

Two video cameras were set up in the room to record the foods selected by each subject. The subjects were told that we were observing people's behavior while waiting in lines. No mention was made about any later memory test of the foods.

Four to fourteen days after the foods were eaten, subjects were interviewed individually with either a standard (no mnemonic instructions) interview or the cognitive interview. The retention intervals were approximately the same for the two conditions. For the no-instruction condition, the retention intervals ranged from 4 to 14 days, with a mean of 7.77 days; for the cognitive interview condition, the retention intervals ranged from 4 to 14 days with a mean of 7.15 days.

All interviews began by asking the subjects to describe all of the foods in as much detail as possible. In the no-instruction interview, subjects were asked to recall the foods that they had selected and also the other foods that were on the table, but which they did not select. Other than this open-ended request, no additional cues were provided to assist the subjects in their recall. After completing their responses, the subjects were asked if there were any more foods they could remember.

In the cognitive interview, subjects were asked initially to re-create the context (environmental and psychological) of the previous eating session and then to generate a complete description of all the foods they remembered, indicating whether they were selected or not. Subjects were asked to recall the items in sequential order, beginning at one end of the table, and then to recall the foods in reverse order by beginning at the other end of the table (varied retrieval). They were asked to take someone else's perspective, such as the person next to them in line, and attempt additional recall from that perspective (extensive recall). Subjects were asked to think about the foods when they initially selected them and also when they sat down to eat (multiple representations). During the interview, subjects were instructed that memory requires intense concentration and that the experimenter expected them to work diligently at the task (focused retrieval). If subjects gave a nondiscriminating response (e.g., "I drank some soda"), they were asked to describe it in more detail. The subjects were encouraged to go through several mental operations (e.g., context reinstatement, multiple representations) to facilitate this process. A model interview is presented in the Appendix to exemplify the principles outlined above. The interview presented was formed by excerpting parts of two interviews conducted during the actual study. For economy and readability, noninformative comments have been edited out of the model interview and the language has been stylized.

Immediately following the interview, subjects were given a list (menu) of fifty-four foods, of which thirty-four were presented at the original meal. The remaining twenty were foils, foods not presented at the meal. Of the foils fifteen were in the same food categories as the presented items (e.g., other beverages, other cheeses) and five were from nonpresented food categories (meats, fish). The test foods were ordered randomly on the sheet. The subjects were asked to respond to each test food, identifying it as either a food that (a) they selected, (b) was available, but which they did not select, or (c) was not available. If unsure, subjects indicated "don't know."

After the final test, subjects were debriefed about the purpose of the experiment and were asked about their reactions. None of the subjects reported that they had suspected a later memory test.

Results

The interviews were audiotape-recorded and later scored for accuracy of recall by comparing the subjects' responses with the videotape of the original meal. All interviews were scored for recall of (a) foods selected[1] (b) foods

1. We refer to the named foods as "selected," as opposed to "eaten," because our videotapes allow us to examine only which foods were selected, not which were actually eaten. Our observations of the subjects' eating behavior suggest that most subjects ate all of the foods selected.

available but not selected, and (c) intrusions (foods not included in the experiment, but which were reported). Responses were scored using both a stringent and a lenient criterion. With the stringent criterion, responses were considered correct only when sufficient detail was provided to discriminate the specific item from all of the other foods available. For example, as there were both green and red grapes, the response, "I ate some red grapes," was considered correct, but "I ate some grapes" was not. However, since there was only one type of olive (green), the response, "I ate some olives," was considered correct. With the lenient criterion, responses that lacked discriminating detail, but which referred to the general class of food (e.g., "some grapes"), were considered correct. Since the patterns of results were similar for the two criteria, we report the results of only the stringent criterion.

In actual field investigations, public-health officials typically conduct food-consumption histories about the suspect meal two to seven days after the meal, shortly after the symptoms are noticed. The retention intervals in the present study, however, ranged from four to fourteen days. In order to examine whether our longer retention intervals might have biased the results, we conducted an initial test to compare the results of those subjects interviewed within the first seven days with those interviewed between eight and fourteen days after the meal. The patterns of results were almost identical for the two groups, other than a slight overall reduction of scores for the longer retention interval. The presented results, therefore, reflect all of the collected data, collapsed across retention interval.

As seen in table 1, more than twice as many foods were recalled with the cognitive interview (12.22) than were with the no-instruction interview (5.92), $F(1, 24) = 24.51, MSe = 5.276, p < .01$. This held equally for selected and available-but-unselected foods; the interaction of interview and food type was not statistically significant, $F(1, 24) = .81, MSe = 4.673, p > .05$.

Although subjects were assigned randomly to the two groups, those in the cognitive interview group selected slightly more foods (8.15) than did those in the no-instruction group (6.00). As a result, there were potentially more selected foods to be recalled for the cognitive interview group than for the

Table 1 Number of Selected and Available-but-Unselected Foods Recalled.

	Interview	
Food	Cognitive	No-Instruction
Selected	5.30	2.69
Available-But-Unselected	6.92	3.23
Total	12.22	5.92

no-instruction group. It is unlikely that this accounts for much of the superiority of the cognitive interview group, since they also recalled more of the available-but-unselected foods, even though they had concomitantly fewer of these foods (25.85) than did the no-instruction group (28.00). A second way to circumvent the problem is to score the data in terms of proportion of selected foods that were recalled. The results are similar to the absolute number recalled. The cognitive interview group remembered a greater proportion of the selected foods (.65) than did the no-instruction group (.45) and also a greater proportion of the available-but-unselected foods (.27 versus .12).

There were forty-five cases in which the subjects recalled inappropriate foods (intrusions). Of these, eight were intraexperimental intrusions, in which subjects indicated that they had selected foods that were available, but which they had not actually selected. The bulk of the intrusion errors ($N = 37$) were extraexperimental, recalling foods that were not available in the experiment. Of these thirty-seven, all were categorical intrusions, recalling foods from food groups that were represented in the experiment (e.g., recalling "mozzarella," as there were other cheeses available). There were no noncategorical intrusions (recalling foods from food groups that were not represented, e.g., saying "salami," given that there were no meat products in the experiment). The average number of intrusions was approximately the same for the cognitive interview (1.77) as for the no-instruction group (1.69), $t\,(24) = .16$, $SE = .335$, $p > .05$. If anything, given that more responses were generated with the cognitive interview, the intrusion *rate* was slightly lower for the cognitive interview (13%) than for the no-instruction group (23%).

The recognition responses (indicating which of the menu items were selected, available but not selected, not available, or "don't know") were scored as follows. A response was scored as a *hit* if the subject indicated that a selected test food was, in fact, selected. A *miss* was scored if any other response (available-but-unselected, not available, or "don't know"[2]) was given. The hit rate was slightly higher for the cognitive interview group (.83) than for the no-instruction group (.79); however, the difference was not statistically significant, $t\,(24) = .14$, $MSE = .042$, $p > .05$. It is likely that this lack of difference was caused by the contribution of a few subjects in the no-instruction group who selected very few foods,[3] as the recognition task

2. "I don't know" responses were scored as if the subject had responded that the food was not available. In a secondary, unreported analysis we omitted these test items and scored only those items to which the subject made a positive response ("selected," "available-but-unselected," or "not available"). The pattern of results was unchanged.

3. In the no-instruction group one person selected only one food, one selected two foods, two people selected three foods, and two selected four foods. In the cognitive interview group only one person selected fewer than five foods.

becomes easier when there are fewer foods to remember. To circumvent this problem, we reanalyzed the data, this time including only those subjects who selected at least five foods. The reanalysis compares seven subjects in the no-instruction group (mean number of foods selected = 8.57) with twelve subjects in the cognitive interview group (mean = 8.58). The difference in hit rates is now somewhat larger, with the cognitive interview group (.83)[4] being marginally greater than the no-instruction group (.75), t (17) = 1.37, SE = .044, $.05 < p < .10$.

One possible explanation for the higher hit rate is that the cognitive interview may simply have lowered people's threshold for saying "selected" to anything, whether or not the food was actually selected. In order to examine this possibility, we also scored the data for false alarms, saying "selected" to nontarget foods (either available-but-unselected or unavailable). The false-alarm rate (likelihood of saying "selected" to a nontarget) was, if anything, slightly lower in the cognitive interview group (.03) than in the no-instruction group (.05). Thus, the cognitive interview group correctly recognized more selected foods and made fewer errors on the nonselected foods.

One method to combine the hit and false-alarm rates into one score, which better reflects overall recognition performance, is to use a signal-detection analysis (Murdock, 1982). When all of the subjects in the two groups were compared, the resulting d' score was slightly, although nonsignificantly, higher for the cognitive interview group (mean d' = 3.30) than for the no instruction group (d' = 3.10), t (24) = .45, SE = .314, $p > .05$.[5] When the scores were reanalyzed using only those subjects who selected at least five foods, the differences were larger (mean d's for cognitive interview and no-instruction groups were 3.34 and 2.39, respectively) and attained conventional levels of statistical significance, t (17) = 2.01, SE = .284, $p < .05$.

The cognitive interview took considerably longer to conduct (15.2 minutes) than did the no-instruction interview (1.5 minutes). To what degree was the superiority of the cognitive interview group due simply to their taking more time to recall the foods? It is difficult to separate the effects of instructions from interview time in the present experiment, as they were so highly correlated with one another ($r_{xy} = .809, p < .01$). Nevertheless, there are some suggestive data from the present experiment that lead us to believe that time itself was not the critical difference between the cognitive and no-instruction groups. First, there was little correlation between interview

4. The group means are weighted scores, so that deleting the data for one subject who selected only three foods had a negligible effect on the group mean.

5. Signal-detection analysis does not permit probabilities of 1.0 or .0; therefore, we used the accepted practice of assigning values of .99 for those subjects who recognized all (p = 1.0) of the selected foods and .01 for those who made no false alarms (p = .0).

time and number of correctly recalled foods within each of the cognitive interview and no-instruction conditions (r_{xy} = .297 and .332, respectively, both $ps > .20$), although note that these low correlations may reflect restricted ranges of interview times within each group. Second, all interviews were terminated when the subjects indicated that they could not recall any more foods, that is, when additional time would be of no value. We defer additional thoughts about the role of interview time until the discussion.

Discussion

The results clearly demonstrate the effectiveness of the cognitive interview as a memory enhancer. More than twice as many foods were recalled with the cognitive interview than with the no-instruction interview, with no more intrusion errors. Comparable, although not as large, effects were found with the recognition test (selecting from a menu). There were more positive identifications (hits) of foods selected and fewer incorrect identifications (false alarms). In summary the cognitive interview supports better recollection with no obvious cost, except for the additional time. Decker et al.'s (1986) pessimism about the unlikelihood of improving food-consumption histories with better interviewing procedures, fortunately, seems to be unfounded.

It is important to note that the cognitive interview is made up of several component principles (e.g., context reinstatement, focused retrieval, multiple representations), each of which is effected by a variety of interviewing techniques. For example, focused retrieval is effected by (a) asking open-ended questions, (b) not interrupting respondents during a response, (c) minimizing extraneous noises, and so on. Because all of these techniques were used as a package in the cognitive interview, we cannot isolate the effect of any one specific technique. It may be that in the present experiment the entire benefit of the cognitive interview was due to only one of the cognitive principles or only one of the various interviewing techniques. Perhaps the superiority of the cognitive interview reflects only that subjects were encouraged to retrieve the events in different ways. Similarly, one can argue that perhaps the entire effect was due to the context reinstatement component. A similar argument can be made about any of the individual components, either at the level of cognitive principles or specific interviewing techniques. We cannot refute this argument on the basis of the present study, because, technically, all of the component cognitive principles and interviewing techniques were confounded within the package of the cognitive interview. We have reason to believe from other studies, however, that each of the component principles enhances recollection. For example, the effect of context reinstatement has been demonstrated by Tulving and Thomson (1973), and of varied retrieval by Anderson and Pichert (1978). It seems reasonable that the relative value of each of the component techniques will vary from one interviewing situation to the next, for instance, from recall to recognition, or from mail surveys to face-to-face interviews. We encourage other interested researchers to explore these interactions by parsing the cognitive interview into its component techniques and examining their relative value as the information-processing requirements of the task change (Fisher, Geiselman, and Amador, 1989).

How valuable is the cognitive interview compared to simply conducting a recognition test? On the surface it appears that the recognition task elicited more information from the no-instruction group than did the recall task for the cognitive interview group. Subjects in the no-instruction condition recognized approximately 79% of all the selected foods; by comparison, subjects in the cognitive interview group recalled only 65% of the selected foods. These numbers are somewhat misleading for three reasons. First, in the present study the recognition test always followed a recall test. Such a double-testing procedure elevates performance on the second (recognition) test, since the earlier attempt to recall strengthens the information (see Richardson, 1985, for a review). A true recognition score, one not preceded by the recall test, would likely have been lower than that observed here. Second, there were many more false recognitions (2.31 false alarms per subject in the no-instruction group) than false recalls (1.77 intrusions per subject in the cognitive interview group), so that the recognition task generated more misleading clues than did the cognitive interview. Third, and most important from the practical aspect, in order to construct a comprehensive recognition test, with all of the target items present, one would have to know the target items prior to constructing the test. In the laboratory this does not present a problem, since the experimenter knows the target items. In the field, however, where the interviewer does not know the target items prior to the interview, it would be difficult to construct such a comprehensive recognition test.

In the present experiment the cognitive interview took considerably longer to conduct than the no-instruction interview. To what extent does this extra time account for the superiority of the cognitive interview? An implicit assumption behind this question is that time, per se, affects recollection; that is, the more time one spends in retrieval, the better will be recollection. We are not aware of any empirical evidence to support this claim. The most relevant study is one by Roediger and Thorpe (1978), who showed in a free-recall task that subjects generated more correct responses as the allotted response time increased; however, this came at the expense of increased intrusions. Thus, the additional response time did not enhance recall as much as it lowered the threshold for responding. Note that in the present study the increase in number of correct items recalled by the cognitive interview group was not accompanied by an increase in the error rate. On the contrary, the intrusion rate was, if anything, lower in the cognitive interview group (13%) than in the no-instruction group (23%). Second, in the present study, length and type of interview were highly correlated, thereby effectively precluding a separation of the two factors. Other studies, however, have shown that the superiority of the cognitive interview is not attributable to its greater length. In a recent laboratory study by Fisher and Layton (1990), interview time was experimentally controlled. Nevertheless, recall of subject-performed actions was better when tested with the cognitive interview than with a no-instruction format. In the one field test documenting the technique's effectiveness, police detectives took no longer to conduct the cognitive interview than a standard police interview (Fisher, Geiselman, and Amador, 1989). We are confident, therefore, that the cognitive interview enhances recall because of the mental operations it promotes, not because of the additional time that may be required.

Assuming it does take somewhat longer to conduct the cognitive interview, how impractical is that for survey researchers? That depends in part on the time demands of the particular survey. If a very large number of respondents must be interviewed and there are stringent time constraints, then the additional time may preclude using the cognitive interview. If the number of respondents is relatively small, however, as in the outbreak of legionnaires' disease, then the extra time associated with the cognitive interview is not extremely debilitating. Some surveys with large numbers of respondents proceed in two stages. In the first stage an intensive face-to-face interview is conducted with a few target individuals to determine the range of potential responses. For example, if a foodborne outbreak occurred in a large hotel, intensive interviews might be conducted with some of the waiters, chefs, or other kitchen personnel to determine the range of foods that were available at the target meal. In the second stage of the investigation, the named foods would then be presented in a recognition format to all of the hotel's guests to determine who ate which foods. In such cases the cognitive interview would be most appropriate for the first stage, where ample time is available for a few intensive interviews. The alternative, of eliminating the first stage and simply presenting a comprehensive list of foods for respondents to recognize, has the potential drawback of taxing the respondent with such a long list of choices that the questionnaire becomes unreliable.

It has been shown across a wide variety of surveys that not only the number but also the specific alternatives provided can alter the respondent's choice and the user's interpretation of the response (see Schwarz, 1990, for a review). Given the importance of selecting the appropriate set of alternatives when constructing the survey instrument, it may be worthwhile initially to conduct a few intensive interviews with a representative sample of individuals to determine the ultimate set of alternatives. Here, too, in the preliminary stages, when more intensive interviews can be conducted, the cognitive interview might be of greatest value.

Whereas the present study examined the effectiveness of the cognitive interview in recalling the details of an isolated event, many survey researchers are interested in recalling events that occur repeatedly, for instance, "How often did you visit a physician during the past year?" A recent study by the National Center for Health Statistics (1989) showed that an effective method to elicit such information is first to decompose the series of events into specific isolable events (e.g., the last) and probe for details of each specific event. In the second phase the respondent is encouraged to construct a personal time line to determine whether the recalled event falls within the specified limit (one year). We suspect that a modification of the presently described cognitive interview can be useful in the initial decomposition phase. For example, the respondent can be encouraged to use varied retrieval operations by probing along different dimensions. One can probe the various events temporally ("Can you remember any specific visits to the doctor during the afternoon? the morning?"), spatially ("Can you remember any examinations in the closest (farthest) examination room?"), proprioceptively ("Can you remember any appointments where you had to stand (sit) most of the time?"), emotionally ("Can you remember any examination in which you were particularly happy (upset) with the results?"), and so on. As a second suggestion, respondents can be encouraged to reconstruct their mental operations associated with specific appointments ("Think about speaking with the receptionist to make an appointment. Did you try to schedule your appointment around any other events? If so, which events?"). We leave it to other investigators to determine how the cognitive interview can be adapted to meet their specific needs.

One immediate application of the cognitive interview for public-health workers is in the investigation of sexually transmitted diseases, especially when respondents have had contact with many sexual partners, so that recollection becomes problematic. From a theoretical perspective, somewhat different memory processes may be used in investigating the two types of event, as the target information (names of foods versus action sequences) differs (see Cohen, Peterson, and Mantani-Atkinson, 1987). As such, the cognitive interview will have to be modified to accommodate the new task's demands. Nevertheless, it may serve as the basis of an effective interviewing technique.

How useful is a technique like the cognitive interview for improving survey research in general? We suspect that it depends in large part on the degree of direct interaction between the respondent and the questioner. At one extreme, the written mail survey, there is virtually no direct interaction between the questioner and the respondent; at the other extreme, the face-to-face interview, there is maximum interaction between the two; and in the middle, the telephone interview, there is an intermediate degree of interaction. Our experience is that the cognitive interview will be more effective as the degree of interaction increases between questioner and respondent. Some principles (e.g., context reinstatement) can be incorporated into interviews with minimal interaction, for example, by providing written instructions encouraging the respondent to re-create the original context. However, there is little guarantee that respondents will follow these written instructions without being monitored. Other principles are virtually impossible to implement in a noninteractive interview. For example, the principle of focused retrieval is implemented by the interviewer's controlling the interview environment to minimize distractions. Obviously, this is impossible to control in a mail survey and nearly impossible in a telephone interview. To make use of the principle of multiple representations, the interviewer should encourage the respondent to provide an initial open-ended description of the event, so that the interviewer can infer the respondent's various representations of the event. The interviewer probes these various mental representations during the remainder of the interview. This means that the interview questions cannot be predetermined as in a questionnaire. Rather, the interviewer must be flexible enough to tailor the interview to the specific mental representations of the individual respondent. In general we suspect that the cognitive interview will be of only limited value in written questionnaire surveys, but that it will be more valuable in telephone and face-to-face surveys.

The authors would like to thank David Tacher and Irene Praeger for assisting in the data collection. We would also like to thank David Mingay, Brian Cutler, and Paul Blake for their helpful comments on an earlier version of the manuscript.

References

Anderson, R. C., and Pichert, J. W. (1978) Recall of previously unrecallable information following a shift in perspective. *Journal of Verbal Learning and Verbal Behavior* 17, 1–12.

Bryan, F. L. (1973) *Guide for Investigating Foodborne Disease Outbreaks and Analyzing Surveillance Data*. Atlanta, GA: U.S. Department of Health and Human Services.

COHEN, R. L., PETERSON, M., and MANTANI-ATKINSON T. (1987) Interevent differences in event memory: Why are some events more recallable than others? *Memory and Cognition* 15, 109–118.

DECKER, M. D., BOOTH, A. L., DEWEY, M. J., FRICKER, R. S., HUTCHESON, R. H., and SCHAFFNER, W. (1986) Validity of food consumption histories in foodborne outbreak investigations. *American Journal of Epidemiology* 124, 859–862.

FISHER, R. P., and CHANDLER, C. C. (1984) Dissociations between temporally-cued and theme-cued recall. *Bulletin of the Psychonomic Society* 22, 395–397.

FISHER, R. P., and GEISELMAN, R. E. (in press) *Memory-Enhancement Techniques for Investigative Interviews: Increasing Eyewitness Recall with the Cognitive Interview*. Springfield, IL: Thomas.

FISHER, R. P., GEISELMAN, R. E., and AMADOR, M. (1989) Field test of the cognitive interview: Enhancing the recollection of actual victims and witnesses of crime. *Journal of Applied Psychology* 74, 722–727.

FISHER, R. P., GEISELMAN, R. E., and RAYMOND, D. S. (1987) Critical analysis of police interview techniques. *Journal of Police Science and Administration* 15, 177–185.

FISHER, R. P., GEISELMAN, R. E., RAYMOND, D. S., JURKEVICH, L., and WARHAFTIG, M. L. (1987) Enhancing enhanced eyewitness memory: Refining the cognitive interview. *Journal of Police Science and Administration* 15, 177–185.

FISHER, R. P., and LAYTON, R. (1990) Recall of subject-performed actions. Manuscript. Miami, FL: Florida International University.

FISHER, R. P., and QUIGLEY, K. L. (1988) The effect of question sequence on eyewitness recall. Manuscript. Miami, FL: Florida International University.

GEISELMAN, R. E., FISHER, R. P., MACKINNON, D. P., and HOLLAND, H. L. (1985) Eyewitness memory enhancement in the police interview: Cognitive retrieval mnemonics versus hypnosis. *Journal of Applied Psychology* 70, 401–412.

KAHNEMAN, D. (1973) *Attention and Effort*. Englewood Cliffs, NJ: Prentice-Hall.

MANN, J. A. (1981) A prospective study of response error in food history questionnaires. *American Journal of Public Health* 70, 401–412.

MURDOCK, B. B. (1982) Recognition memory. In PUFF, R. (ed.). *Handbook of Research Methods in Human Memory and Cognition*. New York: Academic Press, pp. 1–26.

NATIONAL CENTER FOR HEALTH STATISTICS: MEANS, B., NIGAM, A., ZARROW, M., LOFTUS, E. F., and DONALDSON, M. S. (1989) Autobiographical memory for health-related events. *Vital and Health Statistics*, series 6, no. 2. Public Health Service. Washington, DC: U.S. Government Printing Office.

RICHARDSON, J. T. (1985) The effects of retention tests on human learning and memory: An historical review and an experimental analysis. *Educational Psychology* 5, 85–114.

ROEDIGER, H. L., and PAYNE, D. G. (1982) Hypermnesia: The role of repeated testing. *Journal of Experimental Psychology: Learning, Memory, and Cognition* 8, 66–72.

ROEDIGER, H. L., and THORPE, L. A. (1978) The role of recall time in producing hypermnesia. *Memory and Cognition* 6, 296–305.

SCHWARZ, N. (1990) Assessing frequency reports of mundane behaviors: Contributions of cognitive psychology to questionnaire construction. In HENDRICK, C., and CLARK, M. S. (eds.). *Research Methods in Personality and Social Psychology.* Newbury Park, CA: Sage, pp. 98–119.

TULVING, E. (1974) Cue-dependent forgetting. *American Scientist* 62, 74–82.

TULVING, E., and THOMSON, D. M. (1973) Encoding specificity and retrieval processes in episodic memory. *Psychological Review* 80, 352–373.

APPENDIX

***Model Cognitive Interview between Interviewer* (Int) *and Respondent* (Resp)**

Note: New information is printed in UPPERCASE.

Int: I'd like you to try to put yourself back in the same frame of mind as the night when you walked into the room and tell me what you were thinking or feeling.

Resp: Well, we had just taken a test. . . . I think having the food was a nice way to end the evening after a difficult test.

Int: After you came into the room, what is the first thing you did?

Resp: I signed my name in the notebook that was there, then I looked at the video camera and said my name. Then I picked up a plate and started walking along the tables.

Int: Try to put yourself back to where you were standing at the beginning of the tables, holding your plate. Try to visualize the foods that were on the tables. It may help if you close your eyes . . . (pause) . . . Now, start with the first item on the table and work your way down the table, describing each item to me in as much detail as possible. Try not to leave anything out.

Resp: Next to the plates and utensils were some vegetables. I remember BROCCOLI, CUCUMBERS, TOMATOES, MUSHROOMS. . . . After the vegetables, there were some dips. I think there was a BLUE-CHEESE DIP. There were some NUTS. And I remember getting excited because there was chocolate, CHOCOLATE-COVERED PEANUTS . . .

Int: OK, let's go back to the beginning of the tables again. Close your eyes and try to remember any color of items you might have seen.

Resp: Well, the broccoli was green, the cucumbers were green, and I think there were some GREEN GRAPES. I also remember seeing something orange. It seems like there were some CARROTS.

Int: Let's take this from a different perspective. Can you start at the other end of the table and work in a backward direction? Again, please describe each item in as much detail as possible.

Resp: At the end of the table were two or three bottles of SODA with some cups and ice. Next to the sodas were the chocolates. I remember the chocolate-covered peanuts and M & M's, and I remember chocolate kisses, HERSHEY'S KISSES . . .

Int: After you selected the items from the buffet tables, what did you do?

Resp: I looked around the room and saw two girls from my class, so I walked over to where they were sitting and sat down and started to eat.

Int: Now, put yourself back in your seat. It may help if you close your eyes again and try to visualize your plate. Tell me everything you remember seeing on your plate.

Resp: Starting at 9:00, there was the broccoli. Next to the broccoli was the mushrooms, then the blue-cheese dip, then I think I had some CHEESE . . .

Int: Do you remember what type of cheese it was?

Resp: Yes I think it was CHEDDAR CHEESE . . . then next to the cheese was the chocolate-covered peanuts.

Int: Do you remember what anyone sitting near you was eating?

Resp: I think one of the girls sitting next to me was eating PRETZELS.

Int: Now I'm going to go over all of the items that you told me were on the tables. Try to visualize the tables again and tell me if I leave anything out or if you remember any other items as we go along. . . .